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Project Title:	A Study Of Dietary Retinoid For Prevention Of 56Fe-Induced Cancers In Rat Skin		
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	Earlier carcinogenesis experiments have shown that very high energy, high-Z, particulate radiation (56Fe) in either single dose or multiple dose format is about as carcinogenic in rat skin as low-Z particulate radiation with the same linear energy transfer (LET). An additional finding of these studies was that the cancers and benign tumors could be substantially (~50%) prevented by a non-toxic (250 ppm) level of dietary vitamin A acetate. Preliminary gene expression microarray analysis demonstrated that the retinoid interfered with up-regulation of several radiation-induced genes associated with de-differentiation and increased cell proliferation. The current proposal will follow-up these initial findings by testing the hypothesis that retinoids reduce the carcinogenic action of ionizing radiation by altering the DNA repair-related proliferative response of the pluripotent stem cells, which are considered to be plausible target cells for the carcinogenic action of radiation. Carcinogenesis experiments will evaluate the efficacy of cancer prevention when the dietary retinoid is available during stem cell regeneration. Pluripotent stem cells will be activated via radiation-induced epithelial denudation, and the action of a retinoid on epidermal gene expression will be evaluated via		

Task Description:	commercial and custom gene expression arrays. Microarray analysis will be used to analyze in vivo transit amplifying keratinocytes, in vivo pluripotent stem cells and in vitro keratinocytes (which have stem-like properties). The cyclin D1-mediated proliferation pathway and the p53-dependent DNA repair pathways will be studied in response to the 56Fe ion beam in the presence and absence of a dietary retinoid. The proposed experiments, if successful, will provide 1) better quantitative data than now exists on how cancer risk estimates are affected by short-term versus long-term treatments with a dietary retinoid; 2) a better definition of how closely gene expression changes in different types of keratinocytes are linked to the cancer preventive action of a dietary retinoid; 3) new data to evaluate if in vivo gene expression patterns, especially of cyclin D1 pathway and p53-dependent repair pathway, in pluripotent stem cells differ from than of transit amplifying keratinocytes; and 4) new data to establish whether keratinocyte cell lines in vitro exhibit gene expression patterns in response to radiation and/or retinoid that more closely mimic stem cells or ordinary keratinocytes in vivo. To understand better how gene expression alterations by dietary retinyl acetate is affecting the occurrence of skin cancer following exposure to 56Fe particle beam. By understanding how a simple dietary vitamin might inhibit or delay the onset of skin cancer induced by a heavy charged particle beam, such as the 56Fe beam at Brookhaven National Laboratory, the efficacy of recommending non-toxic anti-oxidative supplements, including vitamin E and celecoxib and their combinations to protect astronauts unavoidably exposed to the galactic background of heavy particles can be evaluated more rationally and/or quantitatively.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	New results implicate inhibition of inflammation as a probable mechanism underlying the cancer preventive effect of dietary vitamin A acetate. These results show in laboratory animals that a dietary supplement of vitamin A acetate is significantly anti-carcinogenic and that the likely mechanism of action involves blocking the expression of genes in the inflammation or response-to-external-stimuli categories. Studies were designed to capitalize on these initial findings to determine if other anti-oxidative dietary supplements, such as vitamin E or celecoxib, produce equivalent or additive anti-carcinogenic effects in combination with each or with vitamin A. The latter agents affected gene expression very differently than vitamin A. Vitamin E was inactive and celecoxib was more active than the radiation and to some degree mimicked the radiation response. The latter response disqualified celecoxib as a test agent, even though the radiation effect was slightly diminished the magnitude of the effect was insufficient to expect substantial effects on cancer inhibition. About 50% of the neoplasias induced by 56Fe ion radiation (a surrogate for space radiation) can be prevented by dietary vitamin A alone.
	A series of experiments on cancer induction in the skin of rats exposed to various doses, fractions and LET values of different types of ionizing radiation, suggest what might become a feasible way to extrapolate radiation-induced cancer risk to low doses, maybe even including zero dose. Low fractionated doses of both low and high LET radiation in rat skin have produced cancer yields comfortably within the range of predictions extrapolated from the general approach described below. Further validation at a molecular level seems warranted and will be the objective of the proposed work. The approach starts from assumptions as follows: 1) cancer is the result of an indeterminant but large number of cancer-relevant mutations, 2) single mutations in a fairly small set of cancer-relevant genes are sufficient to produce the instability needed to produce the large number of mutations needed and 3) initial, cancer-launching mutations are the result of an interaction based on geometrical proximity between 2 cancer-relevant DNA lesions. These assumptions lead to a relationship between cancer-launching mutations (mutations/animal) or cancer yield (cancers/animal) and radiation dose of the following form: $Y(D,L) = CLD + BD2$, where $Y(D,L)$ is cancer yield at a given point in time, D is radiation dose, L is radiation LET and C and B are empirical constants. The initial cancer-launching mutation leads to more and more mutations as cell divisions (and time) go on until a full featured cancer cell finally occurs. Along the way various preneoplasias may occur because of slight proliferation rate advantages over normal cells. However in the end none of these preneoplasias are directly in the lineage of the first cancer to occur and are somewhat irrelevant to the final outcome. Realistically we are not currently able to detect cancer-launching mutations at the single cell stage, but if we could, the yield relationship would have the same dose and LET dependence as the cancers, although the constants would vary. The first term on the right reflects mutations that originate from 2 lesions within the track of single ionizing particles. The higher the LET the more likely an interaction between 2 lesions would be. The second term reflects DNA lesions that originate independently in different radiation tracks. Each lesion is proportional to dose, so an interaction is proportional to dose squared. The final form is a variant of the familiar ‘quadratic’ expression of dual action theory which has been used successfully to describe a variety of radiobiological responses, including chromosome aberrations and cytolethality. A notable, though not new, element here is the specific inclusion of the linear energy transfer (L) in the linear term. A rationale for extrapolation is derived by differentiating both sides of the quadratic form in the previous paragraph with respect to dose which produces a simpler form more suitable for hypothesis testing shown here as Equation 1: $dY(D,L)/dD = CL + 2BD$. (1) The latter is a straight line with an ordinate intersection proportional to LET and a positive slope if $B > 0$. A neoplasia with $B = 0$ violates the assumption of 2 DNA lesions needed for a mutation, instead 1 DNA lesion is sufficient; a situation that could be called by analogy mono-action. Clearly if cancer induction can be shown to be consistent with these simple forms, extrapolation of cancer risk to zero dose would be straight-forward for a variety of different kinds of radiation, including the energetic, heavy charged particles of space radiation. Work on cancer induction modifiers, including anti-oxidant dietary supplements continue in the context of these ideas. An experiment was completed that relates to whether prolonged exposure to high LET radiation causes cumulative DNA damage or progression in pre-neoplastic cells to occur more rapidly than expected compared to single exposures. A preliminary result reported here indicates that the space radiation surrogate, 56Fe, induces cancer consistent with simple additivity of 4 individual fractions, which also confirms previous results indicating that 2 fractions were additive. Much longer treatment times will be needed to resolve the critical question that bears heavily on cancer risk estimates, namely does irradiation of preneoplastic cells during prolonged exposures accelerate the carcinogenic progression.

Task Progress:	The objective of the research is to investigate how cancer preventive agents, especially anti-oxidative agents like vitamin A, vitamin E and celecoxib (a specific Cox-2 inhibitor) and their combinations stimulate or retard gene expression during cancer development in rat skin irradiated with 56Fe ions. Retinoids have shown chemopreventive activity in several animal models, but the molecular mechanism underlying the inhibition remains to be established. As an initial approach to a better understanding of this mechanism, this project focuses on discovering how a non-toxic, naturally occurring retinoid (retinyl acetate) in the diet alters gene expression after exposure of rat skin to electrons or 56Fe ion radiation. Specifically, the differential display method based on gene chip technology and the BRB-ArrayTools software from NIH is used to identify genes whose expression is 1) altered by radiation alone, 2) or by retinyl acetate alone, 3) or by radiation but reversed by the presence of retinyl acetate. Furthermore, the affected genes are grouped in accordance with the GoTree system that groups cellular functions and identifies possible pathways involved in the process. Results from this work show how retinyl acetate alters gene expression patterns in rat skin upon irradiation at carcinogenic doses of either electrons or 56Fe ions. To the extent that retinoid-responsive genes are involved in altering carcinogenesis, the results have identified genes involved in the broader area of the vitamin A role in skin cancer, including, immune function genes, oxidative stress genes and external stress response genes. Current results indicate the inhibition of immune function and oxidative stress may be the predominant mode of vitamin A action in relation to cancer prevention. The work also evaluates the carcinogenic action of multiple doses of 56Fe ion radiation. Multiple approaches are being employed, including finding optimal cancer preventive doses of a non-toxic dietary retinoid in rats, assaying the effect of high LET ionizing radiation and the retinoid on gene expression patterns in the rat tissue and developing models of carcinogenic action relevant to evaluating human cancer risk on the basis of cancers produced in experimental animals. The animals were exposed to the 56Fe ion beam at the Brookhaven National Laboratory facility and cancer incidence and gene expression patterns are being evaluated at the A.J. Lanza Laboratory of the New York University School of Medicine. The monitoring of gene expression patterns is carried out by using an Affymetrix microarray system, while the monitoring of variables such as cell proliferation rates, amount of DNA damage, etc. are carried out by flow cytometry. The anti-inflammatory activity of vitamin A seems to be its primary mode of action for preventing epithelial cancers and fibromas. Although it causes squamous differentiation, which mighe be anti-carcinogenic as a result of an apoptotic-like mechanism, preliminary results fail to support the notion that apoptosis, per se, is involved in the inhibitory action of vitamin A on skin cancer induction, as genes in the apoptotic pathway are neither inhibited by the radiation nor induceded by the vitamin. In summary, the results presented here indicate that vitamin A, when administered prior and during exposure to ionizing radiation, has the potential to mitigate the carcinogenic activity of the ionizing radiation including space radiation as represented by energetic 56Fe ions. The results of the microarray work associated with this project suggest rather strongly that oxidative stress may be a significant pro-carcinogenic alteration produced by both low and high LET radiation and that vitamin A is about equally effective for preventing cancers induced by both types of radiation. This opens up a new and simple pathway whereby an anti-inflammatory inhibitor may be used to prevent cancer induction by space radiation. Gene expression patterns in rat skin exposed to a beam of 56Fe ions, a surrogate of the high-energy, heavy ion galactic radiation background, have been examined as a basis for obtaining a better understanding of the possible mechanism behind vitamin A's radio-protective activity. At 1 d after 3.0 Gy of 56Fe ion irradiation, the expression of 110 genes were significantly up-regulated ($p <= 0.05$) in comparison to control, while no genes were altered by the VitA supplement alone. Combined with 56Fe ion radiation, the VitA supplement blocked the expression of 88 (80%) of the 110 genes and eliminated 16 of 18 gene categories that were significantly-altered (all increased) by the 56Fe ion radiation. Categories with large numbers of genes eliminated by the retinoid included 'response to stress' – 33 genes, 'response to biotic stimulus' – 38 genes, 'signal transduction' – 35 genes and 'regulation of cellular/physiological process' – 40 genes. Even for 'immune response' and 'response to biotic stimulus', the only 2 categories that remained significantly-altered in the present of the vitamin, the combined number of altered genes was reduced from 74 to 13. No significant gene expression alterations were found at 7 d relative to control. The results indicate that at 1d dietary VitA strongly interfered with 56Fe ion-induced gene expression within broad categories of response-to-stimulus, immune-related and stress-related genes, implying that the latter gene categories likely have a role to play in the radioprotective action of the vitamin.
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